

```

????????????????????????????????????????? */ :root, [data-theme="light"] { --bg-page:
#f8fafc; --bg-card: #ffffff; --bg-muted: #f1f5f9; --bg-header: #f8fafc;
--border: #e2e8f0; --text-1: #0f172a; --text-2: #475569; --text-3: #94a3b8;
--accent: #6366f1; --accent-bg: #eef2ff; --pass-text: #15803d; --pass-bg:
#f0fdf4; --warn-text: #b45309; --warn-bg: #fffbeb; --fail-text: #b91c1c; --fail-
bg: #fef2f2; --shadow: 0 1px 3px 0 rgb(0 0 0 / 0.07); } /* ?? Design Tokens:
Dark ?????????????????????????????????????????????????????????????????? */ [data-theme="dark"] {
--bg-page: #0f172a; --bg-card: #1e293b; --bg-muted: #1e293b; --bg-header:
#1e293b; --border: #334155; --text-1: #f1f5f9; --text-2: #94a3b8; --text-3:
#64748b; --accent: #818cf8; --accent-bg: #1e1b4b; --pass-text: #4ade80; --pass-
bg: #052e16; --warn-text: #fbbf24; --warn-bg: #1c1400; --fail-text: #f87171;
--fail-bg: #1c0505; --shadow: 0 1px 3px 0 rgb(0 0 0 / 0.3); } /* ?? Density
Scale ?????????????????????????????????????????????????????????????????? */ /* Default
(comfortable) spacing: these vars are overridden per density. */ :root { --d-py:
0.75rem; /* table cell vertical padding */ --d-px: 1rem; /* table cell
horizontal padding */ --d-gap: 1rem; /* grid/flex gap */ --d-card: 1.5rem; /*
card padding */ } [data-density="compact"] { --d-py: 0.35rem; --d-px: 0.625rem;
--d-gap: 0.5rem; --d-card: 1rem; font-size: 0.875rem; } [data-
density="spacious"] { --d-py: 1.125rem; --d-px: 1.375rem; --d-gap: 1.5rem;
--d-card: 2rem; } /* ?? Audience Gating
????????????????????????????????????????????????????????????????? */ /* Sections marked .aud-
manager are hidden for the executive audience. Sections marked .aud-engineer are
hidden for executive and manager. */ [data-audience="executive"] .aud-manager,
[data-audience="executive"] .aud-engineer { display: none; } [data-
audience="manager"] .aud-engineer { display: none; } /* ?? Base Styles
????????????????????????????????????????????????????????????????? */ *, *::before, *::after
{ box-
sizing: border-box; } body { background: var(--bg-page); color: var(--text-1);
font-family: ui-sans-serif, system-ui, -apple-system, sans-serif; margin: 0;
padding: 0; transition: background 0.2s ease, color 0.2s ease; } main { max-
width: 64rem; margin: 0 auto; padding: 2.5rem 1.5rem; } /* ?? Controls Bar
????????????????????????????????????????????????????????????????? */ .ctrl-bar { position:
sticky; top: 0; z-index: 20; background: var(--bg-card); border-bottom: 1px

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solid var(--border); padding: 0.5rem 1.5rem; display: flex; align-items: center;
gap: 0.75rem; flex-wrap: wrap; box-shadow: var(--shadow); } .ctrl-brand { font-
size: 0.7rem; font-weight: 700; letter-spacing: 0.15em; color: var(--accent);
margin-right: auto; text-transform: uppercase; } .ctrl-group { display: flex;
align-items: center; gap: 0.35rem; } .ctrl-label { font-size: 0.68rem; text-
transform: uppercase; letter-spacing: 0.08em; color: var(--text-3); } .ctrl-
select, .ctrl-btn { font-size: 0.78rem; padding: 0.2rem 0.5rem; border: 1px
solid var(--border); border-radius: 0.375rem; background: var(--bg-muted);
color: var(--text-1); cursor: pointer; transition: background 0.15s, border-
color 0.15s; } .ctrl-select:hover, .ctrl-btn:hover { background: var(--accent-
bg); border-color: var(--accent); color: var(--accent); } .ctrl-btn--active {
background: var(--accent) !important; color: #fff !important; border-color:
var(--accent) !important; } /* ?? Cards
????????????????????????????????????????????????????????????????????????? */ .card {
background: var(--bg-card); border: 1px solid var(--border); border-radius:
1rem; padding: var(--d-card); margin-bottom: 2rem; box-shadow: var(--shadow); }
.card-title { font-size: 1.125rem; font-weight: 600; color: var(--text-1);
margin: 0 0 1rem; } .card-subtitle { font-size: 0.95rem; font-weight: 600;
color: var(--text-1); margin: 1.25rem 0 0.625rem; } /* ?? KPI Grid
????????????????????????????????????????????????????????????????????????? */ .kpi-grid {
display: grid; gap: var(--d-gap); grid-template-colum
ns: repeat(auto-fill, minmax(9rem, 1fr)); margin-top: 1.25rem; } .kpi-cell {
background: var(--bg-muted); border-radius: 0.75rem; padding: var(--d-py)
var(--d-px); } .kpi-label { font-size: 0.68rem; text-transform: uppercase;
letter-spacing: 0.1em; color: var(--text-3); margin: 0; } .kpi-value { font-
size: 1.1rem; font-weight: 600; color: var(--text-1); margin: 0.2rem 0 0; } /*
?? Tables ?????????????????????????????????????????????????????????????????? */ .tbl-
wrap { border-radius: 0.75rem; overflow: hidden; border: 1px solid
var(--border); margin-top: 1rem; } table { width: 100%; border-collapse:
collapse; background: var(--bg-card); } thead tr { background: var(--bg-header);
} thead th { padding: var(--d-py) var(--d-px); text-align: left; font-size:
0.7rem; font-weight: 600; text-transform: uppercase; letter-spacing: 0.06em;
color: var(--text-2); } tbody td { padding: var(--d-py) var(--d-px); color:
var(--text-2); border-top: 1px solid var(--border); font-size: 0.875rem; } /*
```

Row-level status tints driven by tr.pass / tr.warn / tr.fail classes. These interact with the CSS token system so they invert correctly in dark mode without any extra JavaScript.

```
tbody tr.pass td { background: var(--pass-bg); color: var(--pass-text); }
tbody tr.warn td { background: var(--warn-bg); color: var(--warn-text); }
tbody tr.fail td { background: var(--fail-bg); color: var(--fail-text); }
```

/* ?? Status Labels

```
/* .status-pass { color: var(--pass-text); font-weight: 700; font-size: 0.7rem; text-transform: uppercase; letter-spacing: 0.08em; }
.status-warn { color: var(--warn-text); font-weight: 700; font-size: 0.7rem; text-transform: uppercase; letter-spacing: 0.08em; }
.status-fail { color: var(--fail-text); font-weight: 700; font-size: 0.7rem; text-transform: uppercase; letter-spacing: 0.08em; }
```

/* ?? Badge

```
/* .badge { display: inline-block; border-radius: 999px; padding: 0.15rem 0.625rem; font-size: 0.68rem; font-weight: 600; letter-spacing: 0.06em; }
.badge-accent { background: var(--accent-bg); color: var(--accent); }
```

/* ??

```
code { background: var(--bg-muted); border-radius: 0.25rem; padding: 0.1em 0.35em; font-size: 0.85em; color: var(--accent); }
```

BLSN Theme ? Light ? Dark ? Auto

Density Compact Comfortable Spacious Audience Executive Manager Engineer ? Wave 15 ? No Drift or Regression Detected Drift detection active. BLSN Synthetic Validation Report Theme light · Density comfortable · Audience engineer Vehicle P3 Orion Cooling Mode

helium_closed_loop Nominal Flow 110 g/s Limits 80?140 g/s Sanity Check Telemetry Status Message PASS [PASS] PASS | Nominal mass flow is bounded by configured limits System Verification: Claimed vs. Actual Claimed Limits: 0 | Actually Tested: 0 ? STATUS: UNKNOWN Total Runtime 0.00s Pass Ratio 0.0% Fail Ratio 0.0% Skip Ratio 0.0% Tuple Execution Telemetry Targets (seconds)

Metric	Target	Warn	Fail
ci_total_runtime_s	120.0	180.0	240.0
pytest_runtime_s	15.0	25.0	35.0
pipeline_runtime_s	5.0	10.0	15.0
yaml_json_lint_runtime_s	3.0	6.0	9.0

Analytics Dashboard Executive View Engineering View System Priority & Bottleneck Ranking (Bradley-Terry) Pairwise probability model: $P(i \geq j) = \frac{p_i}{(p_i + p_j)}$ Global Repo Ranking Rank Artifact BT Strength Win Prob vs Top 1 docs/index.html 2.533414 50.00% 2 docs/_includes/metric_card.html 2.317756

47.78% 3 docs/presentation.html 2.309540 47.69% 4
docs/_includes/plotly_render.html 2.246689 47.00% 5 docs/_layouts/default.html
2.161844 46.04% 6 docs/blsn_report.pdf 0.795024 23.89% 7
docs/visuals/plotly_payloads.json 0.481794 15.98% 8
docs/experimental_design_matrix.json 0.417307 14.14% 9
docs/historical_telemetry_ledger.json 0.409257 13.91% 10
docs/federation/federation_cherry_pick.json 0.382788 13.13% Inter-Category
Ranking Rank Category Win Prob
vs Top 1 html 50.00% 2 json 15.19% 3 md 10.52% 4 pdf 6.43% 5 yaml 3.18% 6 yml
2.87% QPLANT Requirements Ranking Bridge Rank Requirement Win Prob vs Top 1
ci_total_runtime_s 50.00% 2 pytest_runtime_s 50.00% 3 pipeline_runtime_s 50.00%
4 yaml_json_lint_runtime_s 50.00% Wave 12: SPC & Telemetry Stats Runtime
Drift 0.0000s Prev Moving Avg 0.0000s Total Runs 2 Success Rate 100.0% DoE
Factors ? A: python_version · B: os_type · C: load_parameter Principal Component
Analysis (PCA Proof) Principal Component Eigenvalue Explained Variance Highest
Weight Variable PC1 0.000000 0.00% total_tests PC2 0.000000 0.00% skip_rate
Metric Markdown Slugs Ci Total Runtime S Pytest Runtime S Pipeline Runtime S
Yaml Json Lint Runtime S Ci Total Runtime S Metric ID: `ci\_total\_runtime\_s`
Target Threshold: **120.0s** Warn Threshold: **180.0s** Fail Threshold:
240.0s Pytest Runtime S Metric ID: `pytest\_runtime\_s` Target Threshold:
15.0s Warn Threshold: **25.0s** Fail Threshold: **35.0s** Pipeline Runtime S
Metric ID: `pipeline\_runtime\_s` Target Threshold: **5.0s** Warn Threshold:
10.0s Fail Threshold: **15.0s** Yaml Json Lint Runtime S Metric ID:
`yaml\_json\_lint\_runtime\_s` Target Threshold: **3.0s** Warn Threshold: **6.0s**
Fail Threshold: **9.0s** Mermaid System Flow Diagram graph LR; SSOT[SSOT Config]
--> PIPELINE[Pipeline Execution]; PIPELINE --> MARKDOWN[Markdown Slug Pages];
PIPELINE --> INDEX[Index HTML]; PIPELINE --> SLIDES[Slides HTML]; INDEX -->
PDF[PDF Export]; /* ?? BLSN Report UI Controls
?? All user preferences are persisted to
localStorage under the "blsn_" namespace so they survive hard reloads without
server round-trips. Theme: light | dark | auto "auto" resolves to light or dark
via prefers-color-scheme and keeps the attribute updated when the OS preference
changes (e.g. switching between light and dark system themes). Density: compact
| comfortable | spacious Controls the --d-py / --d-px / --d-gap / --d-card CSS

```
(audience.engineering || {}).pca_bottlenecks || {}; Plotly.react('plotly-
```

```

telemetry-chart', tel.data || [], patchLayout(tel, themeRaw));
Plotly.react('plotly-pca-chart', pca.data || [], patchLayout(pca, themeRaw)); }
/* ?? Chart-view audience tab toggle ?????????????????????????????? */
window.switchChartView = function (view) { const viewExec =
document.getElementById('view-executive'); const viewEng =
document.getElementById('view-engineering'); const btnExec =
document.getElementById('btn-exec'); const btnEng =
document.getElementById('btn-eng'); if (!viewExec || !viewEng) return; const
isExec = view === 'executive'; viewExec.style.display = isExec ? 'block' :
'none'; viewEng.style.display = isExec ? 'none' : 'block'; if (btnExec)
btnExec.classList.toggle('ctrl-btn--active', isExec); if (btnEng)
btnEng.classList.toggle('ctrl-btn--active', !isExec); /* Trigger resize so
Plotly fills a div that was previously hidden. */ window.dispatchEvent(new
Event('resize')); }; /* ?? Public setters wired to onchange handlers
???????????????????? */ window.setTheme = function (v) { /* Apply the resolved
value to data-theme but store the raw choice (including "auto") so the select
always reflects user intent. */ root.setAttribute('data-theme',
resolveTheme(v)); const el = document.getElementById('ctrl-theme'); if (el)
el.value = v; setPref('theme', v); rerenderPlotly(v); }; window.setDensity = (v)
=> applyAttr('density', 'ctrl-density', v); window.setAudience = (v) =>
applyAttr('audience', 'ctrl-audience', v); /* ?? Initialise from saved
preferences (fall back to SSOT defaults) */ const initTheme = getPref('theme',
'li
ght'); const initDensity = getPref('density', 'comfortable'); const initAudience
= getPref('audience', 'engineer'); /* Apply theme without going through the
setter so the select shows "auto" rather than the resolved value when stored as
"auto". */ root.setAttribute('data-theme', resolveTheme(initTheme)); const
themeEl = document.getElementById('ctrl-theme'); if (themeEl) themeEl.value =
initTheme; applyAttr('density', 'ctrl-density', initDensity);
applyAttr('audience', 'ctrl-audience', initAudience); /* ?? Plotly payload fetch
???????????????????????????????????????????? */ const emptyLayout = () => ({
paper_bgcolor: 'rgba(0,0,0,0)', plot_bgcolor: 'rgba(0,0,0,0)', xaxis: { visible:
false }, yaxis: { visible: false }, annotations: [{ text: 'No Plotly payload
available', showarrow: false, xref: 'paper', yref: 'paper', x: 0.5, y: 0.5, }],

```

```
}); fetch('visuals/plotly_payloads.json') .then((r) => (r.ok ? r.json() :
Promise.reject(new Error('HTTP ' + r.status)))) .then((payload) => { _payload =
payload; rerenderPlotly(getPref('theme', 'light')); }) .catch(() => { const el =
emptyLayout(); Plotly.newPlot('plotly-telemetry-chart', [], el);
Plotly.newPlot('plotly-pca-chart', [], el); }); /* Keep "auto" theme in sync
when the OS preference changes at runtime (e.g. user switches system dark mode
while the tab is open). */ window.matchMedia('(prefers-color-scheme:
dark)').addEventListener('change', () => { const stored = getPref('theme',
'light'); if (stored === 'auto') { root.setAttribute('data-theme',
resolveTheme('auto')); rerenderPlotly('auto'); } }); });
```